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Special section paper

Using mixed methods research designs in health psychology: An illustrated discussion from a pragmatist perspective

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Purpose. To outline some of the challenges of mixed methods research and illustrate how they can be addressed in health psychology research.

Methods. This study critically reflects on the author's previously published mixed methods research and discusses the philosophical and technical challenges of mixed methods, grounding the discussion in a brief review of methodological literature.

Results. Mixed methods research is characterized as having philosophical and technical challenges; the former can be addressed by drawing on pragmatism, the latter by considering formal mixed methods research designs proposed in a number of design typologies. There are important differences among the design typologies which provide diverse examples of designs that health psychologists can adapt for their own mixed methods research. There are also similarities; in particular, many typologies explicitly orient to the technical challenges of deciding on the respective timing of qualitative and quantitative methods and the relative emphasis placed on each method. Characteristics, strengths, and limitations of different sequential and concurrent designs are identified by reviewing five mixed methods projects each conducted for a different purpose.

Conclusions. Adapting formal mixed methods designs can help health psychologists address the technical challenges of mixed methods research and identify the approach that best fits the research questions and purpose. This does not obfuscate the need to address philosophical challenges of mixing qualitative and quantitative methods.

Statement of contribution

What is already known on this subject?

- Mixed methods research poses philosophical and technical challenges.
- Pragmatism in a popular approach to the philosophical challenges while diverse typologies of mixed methods designs can help address the technical challenges.
- Examples of mixed methods research can be hard to locate when component studies from mixed methods projects are published separately.

What does this study add?

- Critical reflections on the author's previously published mixed methods research illustrate how a range of different mixed methods designs can be adapted and applied to address health psychology research questions.

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- The philosophical and technical challenges of mixed methods research should be considered together and in relation to the broader purpose of the research.

In mixed methods research, qualitative and quantitative approaches (or 'components') are utilized together in a single study or series of related studies. Like all research, mixed methods research entails both philosophical assumptions and technical methods of inquiry (Creswell & Plano Clark, 2007). In other words, one can conceptualize mixed methods research as methodology (i.e., a general approach to research that flows from underlying philosophical assumptions) and/or method (i.e., a specific technique for collecting or analysing data; Teddlie & Tashakkori, 2009). Some researchers may be tempted to focus exclusively on the technical challenges of mixed methods research, leading them to see quantitative methods as simply techniques for collecting and analysing numerical data and qualitative methods as simply techniques for collecting and analysing non-numerical (often textual) data. From this perspective, questions about mixed methods become questions about procedures: Which method should be applied first, which participants should be included in which study, how might we integrate the findings from an analysis of semi-structured interviews with the results of a survey? These are important questions and I will return to them, but a purely technical focus encourages a superficial and unsatisfactory approach to mixed methods research. As Wiggins (2011) has shown, focussing solely on technical issues fosters uncritical and un-reflexive practices which result in poor quality research that undermines the potential of mixed methods. Therefore, in this article, I will briefly outline the philosophical challenges of mixed methods research and an increasingly common approach to these ('pragmatism') before focussing on technical challenges and showing how these can be addressed in health psychology.

Philosophical challenges of mixed methods research

The challenges

The philosophical issues around mixed methods research stem from the fact that quantitative approaches are traditionally associated with positivist or (more recently) post-positivist epistemologies, while qualitative approaches are traditionally associated with constructionist or interpretive epistemologies (Creswell, 2003; Guba & Lincoln, 1994; Johnson & Onwuegbuzie, 2004; Lincoln & Guba, 2000). Post-positivist epistemologies typically entail a (realist) belief in an independent reality that is knowable. From this perspective, knowledge is limited only by our technologies of knowing and so bias-free, objective measurement becomes an essential (if ultimately unattainable) goal in the drive to discover universal laws governing behaviour. Constructionist or interpretive epistemologies typically entail a (relativist) belief that the world is only knowable through our conceptual frameworks, which may differ between individuals and cultures. From this perspective, knowledge is inescapably embedded in values and cultures, including the research process itself, and so locally situated and contextualized understandings are sought through inherently subjective means. Because these epistemological differences between qualitative and quantitative approaches influence how we design, conduct, and evaluate research, it is not appropriate to simply combine qualitative and quantitative techniques without attending to epistemological assumptions about knowledge and its legitimate sources (Bryman, 1988; Yardley, 2001). In other words, it is important to maintain the integrity of each component in mixed methods research (Morse, 2003).

However, because of the dominance of quantitative methods in mainstream psychology (Alise & Teddlie, 2010), quantitative researchers keen to employ mixed methods may add qualitative techniques to their studies but employ them from their own implicit (i.e., post-positivist) standpoint (Yardley & Bishop, 2008). For example, researchers may use semi-structured interviews to collect qualitative data about a patient's experience of illness, but view that data as a straightforward account of an individual's illness cognitions and coping strategies, rather than analysing the subjective meanings of illness and how these are constituted through the discursive resources available to the speaker within their broader sociocultural context. While such an approach to qualitative data might be appropriate for some research questions, naïve application of methods in this way is unlikely to optimize the contribution of qualitative methods. The converse is also possible – qualitative researchers may want to follow up an in-depth interview study with a quantitative survey, but unless they address post-positivist concerns around measurement reliability and representative sampling, the survey would not be credible. An appreciation of the deeper differences between qualitative and quantitative methodologies can help avoid such pitfalls and better maximize the potential of qualitative and quantitative components and their combination.

A pragmatist approach

While other successful approaches have been developed and used (e.g., the transformative approach; Mertens, 2010), pragmatism offers an increasingly popular approach to the philosophical challenges of mixed methods research (Bryman, 2006; Cornish & Gillespie, 2009; Dures, Rumsey, Morris, & Gleeson, 2011; Greene & Caracelli, 2003; Morgan, 2014; Tashakkori & Teddlie, 2010; Yardley & Bishop, 2008). In this context, pragmatism should not be equated with a 'practical' or 'expedient' approach (Denscombe, 2008). Instead, adopting a pragmatist epistemology means drawing on pragmatist philosophers such as John Dewey, William James, Charles Sanders Peirce, and Richard Rorty, to develop a more sophisticated approach to mixed methods research (e.g., Cornish & Gillespie, 2009; Johnson & Onwuegbuzie, 2004; Yardley & Bishop, 2008).

Given the diverse positions espoused within the philosophical literature on pragmatism (for a concise historical overview see Hookway, 2013), it is unsurprising that methodologists have developed subtly different interpretations of the meanings of pragmatism for research. Overall, pragmatist approaches to mixed methods research generally acknowledge the epistemological differences between qualitative and quantitative approaches but do not see these forms of inquiry as incommensurable and advocate a shared aim for all research – to produce positive change in the world. Other ways of viewing the world that are generally characteristic of pragmatism and make it appealing for mixed methods research include rejection of objective–subjective dualism, scientific truths as provisional and achievable through diverse sources of experience and experimentation, and knowledge as both constructed and grounded in the world (see Johnson & Onwuegbuzie, 2004; for more general characteristics and limitations of pragmatism). Cornish and Gillespie (2009) and Yardley and Bishop (2008) explain how one interpretation of pragmatism for mixed methods research is to ask not whether the knowledge produced by research accurately represents 'reality' but whether it has valuable external consequences in the context of the researcher's own time and place. For health psychologists, such consequences might include improved quality of life for individual patients with a particular condition or more effective public health services targeting a specific health behaviour. From this perspective, all research should be

evaluated according to the extent to which it achieves its own particular desired external consequences. Addressing method-specific quality criteria in the quantitative and qualitative components of mixed methods research will help achieve the desired external consequences (Yardley & Bishop, 2008); additionally, addressing quality criteria for the overall mixed methods design may also help but such criteria remain contentious (e.g., see Bryman, 2006; Collins, Onwuegbuzie, & Johnson, 2012; Heyvaert, Hannes, Maes, & Onghena, 2013; Pluye, Gagnon, Griffiths, & Johnson-Lafleur, 2009).

Technical challenges of mixed methods research

Design typologies

As explained above, a pragmatist epistemology can frame high-level judgements about mixed methods design (the best design being the one that allows the researchers to achieve their desired external consequences). More technical questions then come to the fore, about how and when to combine specific qualitative and quantitative methods. One popular approach has been to develop typologies of mixed methods designs. As for taxonomies of behaviour change techniques (Abraham & Michie, 2008; Michie *et al.*, 2013), if a single typology of mixed methods designs were to be agreed by consensus, it could offer a common language to facilitate cumulative science, communication, and training. However, at least 15 typologies of mixed methods designs have already been published (Creswell & Plano Clark, 2011). This proliferation may be somewhat daunting, particularly for new researchers, while focussing on designs from one taxonomy alone may be unnecessarily constraining, particularly for complex multiphase projects conducted for multiple purposes (Guest, 2013). One solution is to recognize the value of typologies for simpler projects, say those using two components, while acknowledging the need to go further when designing and describing more complex research (Guest, 2013). I will now review a small selection of published typologies to illustrate some options available to health psychologists and highlight core, recurrent, technical challenges.

In the first edition of their textbook, Creswell and Plano-Clark (2007) proposed four designs for mixed methods research: Exploratory, explanatory, triangulation, and embedded designs. These designs differ along two dimensions: Timing and emphasis. Exploratory and explanatory designs are sequential in that the first component is completed before the second component is begun. Triangulation and embedded designs are concurrent in that the qualitative and quantitative components are typically undertaken simultaneously. Exploratory designs emphasize (i.e., devote greater resources to/give greater weight to the findings during interpretation) the qualitative component, while explanatory designs emphasize the quantitative component. Triangulation designs tend to weight components equally, while embedded designs emphasize either the qualitative or the quantitative component. Figure 1 depicts these designs, which can be modified as necessary, for example, to change the emphasized component or to combine multiple qualitative and quantitative components in more complex research programmes.

Morse and Niehaus (2009) proposed eight mixed methods designs (Table 1). Interestingly, this typology includes four designs that use either qualitative or quantitative methods (not both) and thus does not meet the definition of mixed methods research used here. However, others would include such designs as mixed methods (e.g., Morse, 2010) and they appear relevant to health psychology given recent interest in pluralistic

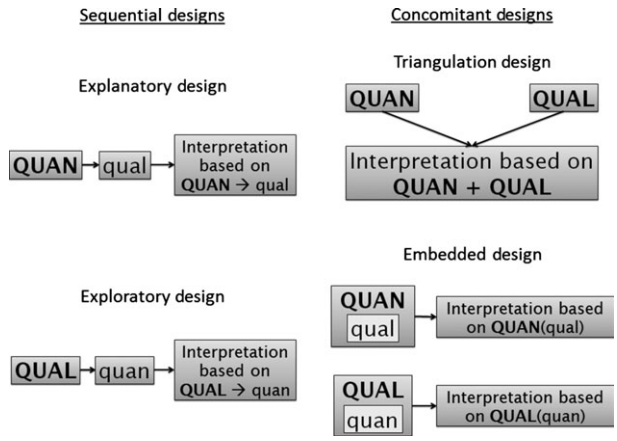


Figure 1. Illustration of four prototypical major mixed methods designs (Creswell & Plano Clark, 2007). QUAN indicates quantitative component; QUAL indicates qualitative component. Capitals indicate component is typically emphasized or prioritized in this design. Lower case indicates component is typically used in a supportive capacity.

Table 1. Simplified summary of eight mixed methods designs (Morse & Niehaus, 2009)

Summary	Core component[a]	Supplemental component[b]	Theoretical drive	Pacing
QUAL + quan	Qualitative	Quantitative	Inductive	Simultaneous
QUAL→quan	Qualitative	Quantitative	Inductive	Sequential
QUAL + qual	Qualitative	Qualitative	Inductive	Simultaneous
QUAL→qual	Qualitative	Qualitative	Inductive	Sequential
QUAN + qual	Quantitative	Qualitative	Deductive	Simultaneous
QUAN→qual	Quantitative	Qualitative	Deductive	Sequential
QUAN + quan	Quantitative	Quantitative	Deductive	Simultaneous
QUAN→quan	Quantitative	Quantitative	Deductive	Sequential

Note. QUAN indicates quantitative component; QUAL indicates qualitative component. Capitals indicate component is the 'core' component. Lower case indicates component is the 'supplemental' component. → indicates one component precedes the other. + indicates components are conducted in parallel.

[a]The core component directs the project overall, providing the theoretical drive.

[b]The supplemental component might enhance, expand on, add description, and/or help explain the core data.

qualitative research (Chamberlain, Cain, Sheridan, & Dupuis, 2011; Frost *et al.*, 2010). Like Creswell and Plano Clark (2007), Morse and Niehaus (2009) distinguished between concurrent (or simultaneous) designs and sequential designs. For Morse and Niehaus, however, any mixed methods design must have a core component and a supplementary component. This contrasts with Creswell and Plano-Clark, who also offered the triangulation design, in which the components are equally weighted.

In the second edition of their textbook, Creswell and Plano Clark (2011) proposed six major mixed methods designs (Figure 2). Compared to their earlier version, this typology

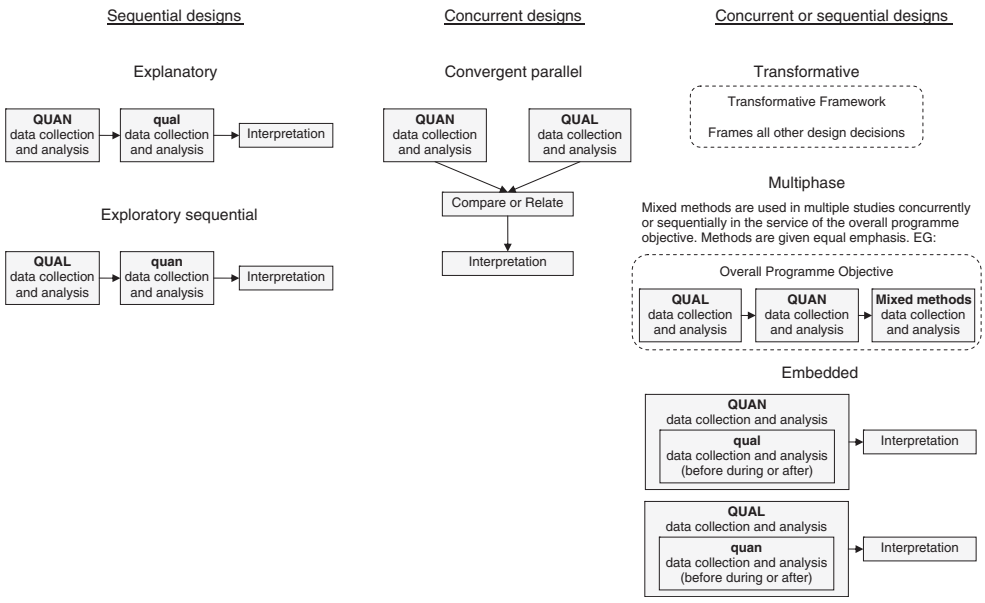


Figure 2. Illustration of six prototypical major mixed methods designs (Creswell & Plano Clark, 2011). QUAN indicates quantitative component; QUAL indicates qualitative component. Capitals indicate component is typically emphasized or prioritized in this design. Lower case indicates component is typically used in a supportive capacity.

presents very similar options for sequential designs but highlights the versatility of the embedded design as usable in a concurrent or sequential fashion. The two new designs added since 2007 are somewhat different. The transformative design describes any combination of methods conducted within an overarching transformative framework, while the multiphase design describes any combination of methods conducted within a more complex programme of research. These additions are thus not highly specified designs in the same sense as the others, but they do remind us that pragmatism is not the only philosophical approach to mixing methods and that more complex programmes of work require more creative designs in which studies using different methods may work together iteratively over time.

In one way or another, all three typologies summarized above explicitly orient to the technical challenges of establishing the respective timing of qualitative and quantitative methods and the relative emphasis placed on each method. Other design typologies also orient to issues of timing and emphasis, for example Morgan's (1998) priority-sequence model constructed four designs based on the order (or sequence) of methods and their relative emphasis (or priority; Table 2); Johnson and Onwuegbuzie (2004) constructed nine designs from every possible combination in a cross-tabulation of emphasis and time order (Table 3). Decisions about timing and emphasis are not made in isolation from considerations of the research's purpose and underlying epistemology. Indeed, Greene's discussions of mixed methods design are organized around five different purposes of mixed methods research: Triangulation – seeking convergence or collaboration using methods with complementary strengths and weaknesses; complementarity – seeking more complete understandings using complementary methods to investigate different

Table 2. Summary representation of four complementary mixed methods designs (Morgan, 1998)

		Priority	
		Principal method: Quantitative	Principal method: Qualitative
Sequence	Complementary method: Preliminary (informs/guides principal component)	qual→QUANT	quant→QUAL
	Complementary method: Follow-up (extends/explains principal component)	QUANT→qual	QUAL→quant

Note. QUANT indicates quantitative component; QUAL indicates qualitative component. Capitals indicate component is prioritized as the principal method. Lower case indicates component is used to complement the principal method. → indicates one component precedes the other.

Table 3. Summary representation of nine mixed methods designs (Johnson & Onwuegbuzie, 2004)

		Emphasis	
		Equal status	Dominant status
Time order	Concurrent	QUAL + QUAN	QUAL + quan QUAN + qual
	Sequential	QUAN→QUAL QUAL→QUAN	QUAL→quan qual→QUAN QUAN→qual quan→QUAL

Note. QUAN indicates quantitative component; QUAL indicates qualitative component. Capitals indicate component is emphasized. Lower case indicates component is de-emphasized. → indicates one component precedes the other. + indicates components are conducted in parallel.

facets of complex phenomena; development – using the first method to inform the development of the second method, for example in terms of sampling frame or questionnaire development; initiation – seeking to uncover paradox or divergence using complementary methods to investigate different facets of complex phenomena; and expansion – using different methods to expand the focus of an investigation into different phenomena (Greene, 2007; Greene, Caracelli, & Graham, 1989). Ultimately, designing mixed methods research is like designing any other form of research: The design must be driven by the research question (Tashakkori & Teddlie, 2010). From this perspective, research design typologies are not best used to select the single 'off-the-shelf' design that approximately fits one's research questions. They may be better used as a source of inspiration to develop tailor-made designs that provide the best possible fit to one's research questions.

Illustrating mixed methods research designs in health psychology

In this section, I will draw on published mixed methods studies conducted with colleagues, to illustrate how different mixed methods designs can be applied in health

psychology research. In these studies, I took a pragmatist stance to mixed methods and aimed to retain the integrity of individual qualitative and quantitative components to maximize their contribution to the overall research goals (Morse, 2003; Yardley & Bishop, 2008). The examples all combine interpretive qualitative methods and post-positivist quantitative methods, but there is no reason why constructionist qualitative approaches (e.g., discourse analysis) could not be included in pragmatist mixed methods studies using these designs.

Using mixed methods to develop and test a new questionnaire

There is a tradition in health psychology of developing new questionnaires by conducting qualitative research to identify relevant constructs/items and following this up with quantitative research that tests the new questionnaire's validity and reliability. Despite not always being labelled as such, this can be considered mixed methods research. However, sometimes quantitative components of questionnaire development studies are emphasized over qualitative components to the extent that the qualitative component may be very narrowly construed and used simply to generate items to assess pre-existing theoretically defined constructs. For example, qualitative methods are often used to generate items for theory of planned behaviour questionnaires (e.g., Francis *et al.*, 2004). This may be satisfactory for some research questions. However, it can artificially limit the insights obtained from the qualitative component as it does not promote the collection of qualitative data capable of critiquing the researcher's pre-existing theoretical constructs: Constructing a theory-based interview topic guide with questions to elicit content for questionnaire items assessing attitudes, subjective norms, and perceived behavioural control is unlikely to elicit talk about other issues that might be important to the participant but do not fit well with these existing constructs.

Conceptualizing questionnaire development as a mixed methods project from the start may encourage more emphasis on the qualitative component than might occur otherwise. For example, I was involved in developing a condition-specific measure of treatment beliefs in low back pain (LBP). This project was designed as a sequential exploratory project (Creswell & Plano Clark, 2007) because our primary aim was to use qualitative methods to explore patients' beliefs and identify common themes, and our secondary aims were to extend this work by designing a new questionnaire to assess our common themes and to test the questionnaire quantitatively. In phase 1, we conducted 13 focus groups with 75 purposively sampled adult primary care patients to explore patients' beliefs about LBP treatments. Thematic analysis (see Joffe & Yardley, 2004) resulted in the identification of four dimensions that participants used to evaluate diverse LBP treatments – credibility, effectiveness, concerns, and individual fit. These dimensions were best understood in the context of participants' broader experiences of seeking treatment, especially their desire for a 'proper' diagnosis that helped them to understand why they had LBP and allowed them to view treatment decisions as rational rather than 'trial and error'. In phase 2, the focus group transcripts were revisited to generate questionnaire items to tap each dimension and contextual theme revealed by the thematic analysis. After a small think-aloud study to refine item-wording, the new questionnaire was tested in 429 LBP patients and analysed using item-response theory and classical test theory to determine its psychometric properties. The qualitative and quantitative components are being published separately (Dima *et al.*, 2013, 2014), which allowed us to fully report each component in its own terms but discouraged a consideration of how the phase 2 results relate back to the phase 1 findings. The

quantitative analysis revealed that our questionnaire measured the four dimensions that we had hypothesized, based on our qualitative data, organized participants' beliefs about diverse LBP treatments. However, it did not effectively distinguish between all of the contextual domains we set out to measure. Apparent discrepancies between qualitative and quantitative components are common in mixed methods studies and can provide additional insights (Moffatt, White, Mackintosh, & Howel, 2006). We view the apparent discrepancy in this project as suggesting that while well-defined unitary constructs can be assessed quantitatively, more complex contextual factors may be best evaluated using qualitative methods.

Using quantitative methods to test ideas derived from qualitative work

Sequential designs in which qualitative components precede quantitative components have other applications in health psychology, beyond questionnaire development. For example, one might conduct a major grounded theory study to develop a novel theory about a poorly understood phenomenon such as public responses to avian flu. This could be followed up with a quantitative study to test hypotheses derived from the new theory. I was involved in a small-scale sequential study which illustrates this general approach. In phase 1, we aimed to explore patients' experiences of acupuncture. A maximum variation sample of 35 patients participated in semi-structured interviews, and inductive thematic analysis explored how patients choose their acupuncturist. Patients experienced this choice as important but not always straightforward and emphasized an acupuncturist's technical competence (e.g., needling skills) and personal attributes (e.g., likeability) as particularly relevant. In phase 2, we designed an experiment to test hypotheses derived from our qualitative findings. Seventy-three participants rated their likelihood of consulting eight fictional acupuncturists, representing all combinations of three-two-level factors (acupuncturist gender, acupuncturist trained in United Kingdom or China, and acupuncturist with or without conventional biomedical qualifications). A 2 (acupuncturist gender) $\times$ 2 (acupuncturist training location) $\times$ 2 (acupuncturist qualifications) $\times$ 2 (participant gender) mixed ANOVA on participants' self-rated likelihood of consulting an acupuncturist confirmed our hypotheses that participants would prefer female acupuncturists and those with conventional biomedical qualifications. In this project, therefore, we were able to use qualitative methods to identify a topic of relevance to patients in the 'real world' and the factors that they perceived as important in their decision-making; the subsequent quantitative study enabled us to test whether these factors did actually influence patients' preferences in a larger sample faced with a hypothetical choice. We published this project in a single paper in a medical journal (Bishop, Massey, Yardley, & Lewith, 2011) which allowed us to describe the mixed methods design but meant we had to present the whole project very concisely and were unable to fully contextualize the qualitative findings. Given that we began this project with an exploratory stance and devoted more resources to the qualitative component, we described it as having a sequential exploratory design. However, the use of hypothesis testing (rather than exploratory statistics) in the quantitative phase might suggest that this label is not entirely satisfactory (see also Guest, 2013). While this design worked well for our relatively small-scale project, it might be more challenging to focus a quantitative study following a larger qualitative study that generated more hypotheses. In some cases, additional steps, for example developing and validating new measures, might be needed in between the two main phases.

Using qualitative methods to explain quantitative results

Sequential designs can also be constructed in which a quantitative component precedes a qualitative component. In sequential explanatory designs (Creswell & Plano Clark, 2007), the qualitative component is employed to try to explain or contextualize the earlier quantitative results. For example, a quantitative health survey could test hypotheses about causal relationships between health beliefs and health behaviours or establish prevalence rates of coping strategies among patients with chronic physical illness. If the health survey revealed a strong but unexpected correlation between beliefs about wearing seat belts and eating habits, then one could sample people based on particular combinations of these variables to take part in qualitative interviews to generate possible explanations. If the prevalence survey found that a particular coping strategy is especially common in one physical condition but not another, then one could include people with both conditions in a follow-up qualitative study. This design can raise ethical challenges as it can be difficult to know before obtaining the quantitative results what the subsequent qualitative study will focus on. If qualitative participants might be sampled from among quantitative participants, then this needs to be planned in advance to obtain consent to recontact people. A new focus for the qualitative component might also necessitate amending protocols and reapplying for regulatory approvals.

To illustrate the sequential explanatory design, I will discuss a mixed methods study of patients choosing osteopaths, conducted with colleagues (Bishop, Bradbury, Jeludin, Massey, & Lewith, 2013). We began with an experimental design administered by postal survey, in which male and female respondents rated their likelihood of consulting eight fictional osteopaths who represented all combinations of three-two–level factors derived from previous literature: Osteopath gender; working in the public sector (i.e., free at point of access) or private sector (i.e., patient pays); and osteopath with or without conventional biomedical qualifications. The results of a 2 (osteopath gender) $\times$ 2 (health care sector) $\times$ 2 (osteopath qualifications) $\times$ 2 (respondent gender) mixed ANOVA on respondents' self-rated likelihood of consulting an osteopath showed that respondents preferred osteopaths with conventional biomedical qualifications, but (contrary to our hypotheses) the other factors were not significant. There was also a notably high rate of missing data (16% of questionnaires were unusable). We then conducted a secondary qualitative analysis of semi-structured interviews with osteopathy patients that explored their use and experiences of osteopathy in depth. However, while this qualitative data helped to contextualize our findings, it did not help us to generate explanations of the non-significant quantitative results, probably because the interviews tended to explore what participants thought was important when choosing osteopaths rather than what was unimportant. More helpfully, the qualitative findings did suggest why the questionnaires had so much missing data: Personal recommendations from trusted others were highly valued by patients choosing osteopaths, but the experiment was unrealistic in this regard as it forced respondents to imagine a situation in which they had no personal recommendations. Overall, it may have been better to conduct a new qualitative study in which the data were collected after obtaining the quantitative results. For example, a think-aloud study would allow us to investigate how people interpreted and responded to our questionnaire and a new qualitative interview study with osteopathic patients could explicitly focus on how they chose their osteopath and explore their views about the specific factors examined in the survey.

Using qualitative and quantitative methods to investigate different facets of the same phenomenon

I was involved in a mixed methods study that collected qualitative and quantitative data simultaneously from the same participants and aimed to evaluate general practitioners' (GPs') use of placebos in routine primary care. We used a web-based questionnaire with closed and open-ended questions to collect data from 783 GPs. The quantitative component measured the prevalence of use of different types of placebos by GPs in routine clinical practice; for example, 12% of GPs reported having used pure placebos such as sugar pills at least once in their career. The percentage of GPs who agreed with various attitudinal statements was also assessed; for example, 52% agreed that it is acceptable to use pure placebos for their psychological effect (Howick *et al.*, 2013). The qualitative component explored GPs' views on placebos, using descriptive analysis to inductively place responses to open-ended questions into three categories: Defining placebos and their effects in general practice; ethical societal and regulatory issues; and reasons why a GP might use placebos and placebo effects (Bishop *et al.*, 2014). While both components addressed the same topic, they did not address a single research question. Instead, they provided complementary insights that together provided a more comprehensive understanding of GPs' use of placebos in clinical practice than would have been achieved by either component alone. This could therefore be considered a complementarity design (Greene *et al.*, 1989). Including a few open-ended questions on an otherwise quantitatively oriented questionnaire may not make the most of qualitative methods' ability to delve into a topic in depth, as it did not allow us to probe GPs for more detailed responses. However, it did allow GPs to express their views anonymously on a contentious and ethically sensitive topic, and obtaining qualitative and quantitative data from the same participants helped to enhance the coherence of this project overall.

Embedding qualitative research in a clinical trial

Typically in embedded (or nested) designs, one component is much larger and more important than the other and the same participants are involved in both. The smaller component is explicitly designed and viewed as supporting the larger component. A common example of this design is the small qualitative study nested within a major clinical trial. Because one component is heavily emphasized (usually but not always the quantitative one, see Donovan *et al.*, 2002 for a counter-example), the supporting component can be somewhat overlooked and/or under-resourced. It may even risk being subsumed within the main component, which could mean it is designed and/or evaluated inappropriately according to quality criteria that better apply to the main component.

I was involved in a qualitative study embedded in a pilot trial of web-based cognitive behaviour therapy (CBT) for irritable bowel syndrome. The trial had a complex factorial design in which participants were randomized to one of three medications (methylcellulose, mebeverine, or placebo) and then they were also randomized to one of three self-management conditions (web-based CBT plus nurse support, web-based CBT only, or no additional input), forming nine groups in total (Everitt *et al.*, 2013). The embedded qualitative study aimed to explore participants' experiences of web-based CBT to identify possible improvements for a future definitive trial. Semi-structured interviews were conducted with participants sampled from each of the web-based CBT conditions. Inductive thematic analysis identified three types of engagement with CBT. One group of participants had limited or no engagement with the intervention, another group engaged with the lifestyle advice and activities, while the third group engaged with the content related to psychological aspects of irritable bowel syndrome (Tonkin-Crine, Bishop, Ellis,

Moss-Morris, & Everitt, 2013). The embedded qualitative study allowed us to suggest improvements to this specific web-based intervention which might also be relevant to other similar interventions, for example emphasizing that the psychological content is relevant to everyone even those who do not currently feel stressed or anxious. Because the qualitative and quantitative studies were so closely related, it was also possible to examine similar issues from different perspectives. For example, the qualitative data revealed some participants who had very limited engagement with the website but who had been categorized quantitatively as compliant with the intervention (based on self-reported usage). This provided convincing evidence that future measures of adherence to the intervention should incorporate more objective meta-data from the website and/or a better self-report measure.

Reflections on the strengths and challenges of different designs

In my experience, sequential designs share common strengths and challenges, while concurrent designs seem to encourage (and lend themselves to) slightly different styles of working. In particular, sequential designs may make it easier to retain the integrity of each method and to evaluate each method according to method-specific quality criteria, because one study is completed before the other begins. Concurrent designs tend to

Table 4. Comparing practical implications of sequential and concurrent mixed methods designs

	Sequential designs	Concurrent designs
Personnel	Can be done by a single researcher or small group of researchers; may appeal to PhD students and trainee health psychologists required to develop qualitative and quantitative research skills (ESRC, 2009; Health Professions Council, 2010)	Easier for a team of researchers because multiple research activities conducted in parallel; plan to manage team-working, group dynamics, and power issues to minimize risk of separating into constituent qualitative and quantitative components (Curry *et al.*, 2012; Lunde, Heggen, & Strand, 2013)
Skills	Challenging for a single researcher to develop the different skills necessary to conduct high-quality qualitative and quantitative research. Training and supervision in qualitative, quantitative, and mixed methods research is needed	Team members can be highly skilled specialist researchers, plus a mixed methods specialist. Can be helpful to have a 'translator' to bridge the different jargon and concepts used by researchers from different methodological traditions
Duration	Take longer to complete	Quicker, but intensive in the short term
Funding	Can be challenging to secure complete funding upfront if not possible to specify second (or subsequent) components before completing first component	Easier to specify all components in advance in sufficient detail for funding applications
Regulatory approvals	Amendments required if cannot specify second (or subsequent) components before completing first component	Easier to specify all components in advance to secure regulatory approvals
Publication	Can be published separately or together, but lend themselves to separate publications as and when each study is completed	Can be published separately or together, but lend themselves to joint publication, as components have closely related research questions and sometimes one component may make little sense in the absence of the other

encourage researchers to blend qualitative and quantitative methods a little more closely, which can make it more difficult to keep in mind the fundamental assumptions, strengths, and limitations of each approach. Conversely, those who are keen to make explicit and strong links between qualitative and quantitative components might find this easier in concurrent designs (in which the links are naturally more apparent) than in sequential designs (which can feel like two separate projects, rather than two components of a mixed methods project). More practical characteristics of sequential and concurrent designs are summarized in Table 4.

Discussion

Mixed methods research involves philosophical and technical challenges. Pragmatism offers a popular set of approaches for addressing the philosophical challenges. Typologies of mixed methods designs offer a smorgasbord of design options and a set of frameworks for thinking through the technical challenges. They also sensitize us to issues which are central to designing more complex mixed methods projects, such as relative emphasis, timing, the purpose of mixing methods, and the stage at which qualitative and quantitative components will be inter-related. Because of the historical dominance of quantitative methods in mainstream health psychology and related disciplines, qualitative researchers may be wary that the mixed methods movement could lead to even more limited use of qualitative methods only in support of quantitative methods (Morse, 2005). Increasing awareness and flexible use of specific mixed methods designs may help to cement mixed methods as the 'third methodological movement' (Teddlie & Tashakkori, 2003), which requires specialist knowledge in itself and which cannot replace stand-alone qualitative (or quantitative) projects.

This paper has briefly reviewed pragmatist approaches to mixed methods research and discussed a selection of design typologies, illustrated the use of different mixed methods designs and reflected on their characteristics. Other important issues that were not discussed include ongoing debates about the integration of qualitative and quantitative methods (Greene, 2008; O'Cathain, Murphy, & Nicholl, 2010; Sandelowski, 2000) and the legitimation and/or quality appraisal of mixed methods research (Bryman, 2006; Collins *et al.*, 2012; Heyvaert *et al.*, 2013; Pluye *et al.*, 2009). Given that health psychologists often research complex real-world problems and accept both qualitative and quantitative methods, we might be well-placed to contribute to such debates. The development of mixed methods research in health psychology can be encouraged by educating students and trainees in mixed methods research as well as in both qualitative and quantitative methods (Onwuegbuzie & Leech, 2005). More published examples of mixed methods research will also help. Publishing qualitative and quantitative components together in a single paper encourages a more detailed consideration of the mixed methods nature of the project, but it can be difficult to do justice to both components; publishing them separately can preclude coverage of mixed methods considerations but allows different components to be targeted to different audiences. One model is to publish components separately in discipline-specific outlets and then also publish an overarching mixed methods paper. As open-access and online publications become the norm, restrictions on length may ease, encouraging publication strategies to be based on more fundamental pragmatist concerns about how to reach the audience most able to implement the knowledge produced by the research.

In conclusion, formal mixed methods designs can help health psychologists think through the technical challenges of mixed methods research and develop an approach

that best fits the research questions and purpose. However, resolving technical challenges should not obfuscate the need to also address philosophical challenges of mixing qualitative and quantitative methods.

References

Abraham, C., & Michie, S. (2008). A taxonomy of behavior change techniques used in interventions. *Health Psychology*, *27*, 379–387. doi:10.1037/0278-6133.27.3.379

Alise, M. A., & Teddlie, C. (2010). A continuation of the paradigm wars? Prevalence rates of methodological approaches across the social/behavioral sciences. *Journal of Mixed Methods Research*, *4*, 103–126. doi:10.1177/1558689809360805

Bishop, F. L., Bradbury, K., Jeludin, N. N. H., Massey, Y., & Lewith, G. T. (2013). How patients choose osteopaths: A mixed methods study. *Complementary Therapies in Medicine*, *21*(1), 50–57. doi:10.1016/j.ctim.2012.10.003

Bishop, F. L., Howick, J., Heneghan, C., Wolstenholme, J., Stevens, S., Hobbs, F. D. R., & Lewith, G. (2014). Placebo use in the United Kingdom: A qualitative study exploring GPs' views on placebo effects in clinical practice. *Family Practice*, *31*(3), 357–363. doi:10.1093/fampra/cmu016

Bishop, F. L., Massey, Y., Yardley, L., & Lewith, G. T. (2011). How patients choose acupuncturists: A mixed-methods project. *Journal of Alternative and Complementary Medicine*, *17*(1), 19–25. doi:10.1089/acm.2010.0061

Bryman, A. (1988). *Quality and quantity in social research*. London, UK: Routledge.

Bryman, A. (2006). Paradigm peace and the implications for quality. *International Journal of Social Research Methodology*, *9*, 111–126. doi:10.1080/13645570600595280

Chamberlain, K., Cain, T., Sheridan, J., & Dupuis, A. (2011). Pluralisms in qualitative research: From multiple methods to integrated methods. *Qualitative Research in Psychology*, *8*, 151–169. doi:10.1080/14780887.2011.572730

Collins, K. M. T., Onwuegbuzie, A. J., & Johnson, R. B. (2012). Securing a place at the table: A review and extension of legitimation criteria for the conduct of mixed research. *The American Behavioral Scientist*, *56*, 849–865. doi:10.1177/0002764211433799

Cornish, F., & Gillespie, A. (2009). A pragmatist approach to the problem of knowledge in health psychology. *Journal of Health Psychology*, *14*, 800–809. doi:10.1177/1359105309338974

Creswell, J. W. (2003). *Research design. Qualitative, quantitative, and mixed methods approaches*. (2nd ed.) Thousand Oaks, CA: Sage.

Creswell, J. W., & Plano Clark, V. L. (2007). *Designing and conducting mixed methods research*. (1st ed.) Thousand Oaks, CA: Sage.

Creswell, J. W., & Plano Clark, V. L. (2011). *Designing and conducting mixed methods research*. (2nd ed.) Thousand Oaks, CA: Sage.

Curry, L. A., O'Cathain, A., Clark, V. L. P., Aroni, R., Fetters, M., & Berg, D. (2012). The role of group dynamics in mixed methods health sciences research teams. *Journal of Mixed Methods Research*, *6*(1), 5–20. doi:10.1177/1558689811416941

Denscombe, M. (2008). Communities of practice: A research paradigm for the mixed methods approach. *Journal of Mixed Methods Research*, *2*, 270–283. doi:10.1177/1558689808316807

Dima, A., Lewith, G. T., Little, P., Moss-Morris, R., Foster, N. E., & Bishop, F. L. (2013). Identifying patients' beliefs about treatments for chronic low back pain in primary care: A focus group study. *British Journal of General Practice*, *63*(612), e490–e498. doi:10.3399/bjgp13X669211

Dima, A., Lewith, G. T., Little, P., Moss-Morris, R., Foster, N. E., Hankins, M., . . . Bishop, F. L. (2014). *Treatment beliefs in low back pain: Development and validation of a condition-specific patient-report measure*. Manuscript submitted for publication.

Donovan, J., Mills, N., Smith, M., Brindle, L., Jacoby, A., Peters, T., . . . Hamdy, F. (2002). Quality improvement report – improving design and conduct of randomised trials by embedding them in qualitative research: ProtecT (prostate testing for cancer and treatment) study. *British Medical Journal*, *325*, 766–769. doi:10.1136/bmj.325.7367.766

Dures, E., Rumsey, N., Morris, M., & Gleeson, K. (2011). Mixed methods in health psychology: Theoretical and practical considerations of the third paradigm. *Journal of Health Psychology*, *16*, 332–341. doi:10.1177/1359105310377537

ESRC (2009). *ESRC postgraduate training and development guidelines 2009*. Swindon, UK: Author.

Everitt, H., Moss-Morris, R., Sibelli, A., Tapp, L., Coleman, N., Yardley, L., ... Little, P. (2013). Management of irritable bowel syndrome in primary care: The results of an exploratory randomised controlled trial of mebeverine, methylcellulose, placebo and a self-management website. *BMC Gastroenterology*, *13*(1), 68. doi:10.1186/1471-230X-13-68

Francis, J. J., Eccles, M. P., Johnston, M., Walker, A., Grimshaw, J., Foy, R., ... Bonetti, D. (2004). *Constructing questionnaires based on the theory of planned behaviour. A manual for health services researchers*. Centre for Health Services Research, University of Newcastle. Retrieved from http://openaccess.city.ac.uk/1735/

Frost, N., Nolas, S. M., Brooks-Gordon, B., Esin, C., Holt, A., Mehdizadeh, L., & Shinebourne, P. (2010). Pluralism in qualitative research: The impact of different researchers and qualitative approaches on the analysis of qualitative data. *Qualitative Research*, *10*, 441–460. doi:10.1177/1468794110366802

Greene, J. C. (2007). *Mixed methods in social inquiry*. San Francisco, CA: Jossey-Bass.

Greene, J. C. (2008). Is mixed methods social inquiry a distinctive methodology? *Journal of Mixed Methods Research*, *2*(1), 7–22. doi:10.1177/1558689807309969

Greene, J. C., & Caracelli, V. J. (2003). Making paradigmatic sense of mixed methods practice. In A. Tashakkori & C. Teddlie (Eds.), *Handbook of mixed methods in social & behavioral research* (pp. 91–110). Thousand Oaks, CA: Sage.

Greene, J. C., Caracelli, V. J., & Graham, W. F. (1989). Toward a conceptual framework for mixed-method evaluation designs. *Educational Evaluation and Policy Analysis*, *11*, 255–274. doi:10.3102/01623737011003255

Guba, E. G., & Lincoln, Y. S. (1994). Competing paradigms in qualitative research. In N. K. Denzin & Y. S. Lincoln (Eds.), *Handbook of qualitative inquiry* (pp. 105–117). Thousand Oaks, CA: Sage.

Guest, G. (2013). Describing mixed methods research: An alternative to typologies. *Journal of Mixed Methods Research*, *7*, 141–151. doi:10.1177/1558689812461179

Health Professions Council (2010). *Standards of proficiency. Practitioner psychologists*. London, UK: Author.

Heyvaert, M., Hannes, K., Maes, B., & Onghena, P. (2013). Critical appraisal of mixed methods studies. *Journal of Mixed Methods Research*, *7*, 302–327. doi:10.1177/1558689813479449

Hookway, C. (2013). Pragmatism. In E. N. Zalta (Ed.), *The Stanford encyclopedia of philosophy*. (Winter 2013 ed.) Retrieved from http://plato.stanford.edu/archives/win2013/entries/pragmatism

Howick, J., Bishop, F. L., Heneghan, C., Wolstenholme, J., Stevens, S., Hobbs, F. D. R., & Lewith, G. (2013). Placebo use in the United Kingdom: Results from a national survey of primary care practitioners. *PLoS One*, *8*(3), e58247. doi:10.1371/journal.pone.0058247

Joffe, H., & Yardley, L. (2004). Content and thematic analysis. In D. F. Marks (Ed.), *Research methods for clinical and health psychology* (pp. 56–68). London, UK: Sage.

Johnson, R. B., & Onwuegbuzie, A. J. (2004). Mixed methods research: A research paradigm whose time has come. *Educational Researcher*, *33*, 14–26. doi:10.3102/0013189X033007014

Lincoln, Y. S., & Guba, E. G. (2000). Paradigmatic controversies, contradictions, and emerging confluences. In N. K. Denzin & Y. S. Lincoln (Eds.), *Handbook of qualitative research*. (2nd ed., pp. 163–188). Thousand Oaks, CA: Sage.

Lunde, Å., Heggen, K., & Strand, R. (2013). Knowledge and power: Exploring unproductive interplay between quantitative and qualitative researchers. *Journal of Mixed Methods Research*, *7*, 197–210. doi:10.1177/1558689812471087

Mertens, D. M. (2010). Transformative mixed methods research. *Qualitative Inquiry*, *16*, 469–474. doi:10.1177/1077800410364612

Michie, S., Richardson, M., Johnston, M., Abraham, C., Francis, J., Hardeman, W., ... Wood, C. (2013). The behavior change technique taxonomy (v1) of 93 hierarchically clustered techniques: Building an international consensus for the reporting of behavior change interventions. *Annals of Behavioral Medicine*, *46*, 81–95. doi:10.1007/s12160-013-9486-6

Moffatt, S., White, M., Mackintosh, J., & Howel, D. (2006). Using quantitative and qualitative data in health services research – what happens when mixed method findings conflict? [ISRCTN61522618]. *BMC Health Services Research*, *6*(1), 28–37. doi:10.1186/1472-6963-6-28

Morgan, D. L. (1998). Practical strategies for combining qualitative and quantitative methods: Applications to health research. *Qualitative Health Research*, *8*, 362–376. doi:10.1177/104973239800800307

Morgan, D. L. (2014). *Integrating qualitative & quantitative methods. A pragmatic approach*. Thousand Oaks, CA: Sage.

Morse, J. M. (2003). Principles of mixed methods and multimethod research design. In A. Tashakkori & C. Teddlie (Eds.), *Handbook of mixed methods in social & behavioral research* (pp. 189–208). Thousand Oaks, CA: Sage.

Morse, J. M. (2005). Evolving trends in qualitative research: Advances in mixed methods design. *Qualitative Health Research*, *15*, 583–585.

Morse, J. M. (2010). Simultaneous and sequential qualitative mixed method designs. *Qualitative Inquiry*, *16*, 483–491. doi:10.1177/1077800410364741

Morse, J. M., & Niehaus, L. (2009). *Mixed method design. Principles and procedures*. Walnut Creek, CA: Left Coast Press, Inc.

O'Cathain, A., Murphy, E., & Nicholl, J. (2010). Three techniques for integrating data in mixed methods studies. *British Medical Journal*, *341*, c4587. doi:10.1136/bmj.c4587

Onwuegbuzie, A. J., & Leech, N. L. (2005). On becoming a pragmatic researcher: The importance of combining quantitative and qualitative research methodologies. *International Journal of Social Research Methodology*, *8*, 375–387. doi:10.1080/13645570500402447

Pluye, P., Gagnon, M. P., Griffiths, F., & Johnson-Lafleur, J. (2009). A scoring system for appraising mixed methods research, and concomitantly appraising qualitative, quantitative and mixed methods primary studies in mixed studies reviews. *International Journal of Nursing Studies*, *46*, 529–546. doi:10.1016/j.ijnurstu.2009.01.009

Sandelowski, M. (2000). Combining qualitative and quantitative sampling, data collection, and analysis techniques in mixed-method studies. *Research in Nursing & Health*, *23*, 246–255. doi:10.1002/1098-240X(200006)23:3<246:AID-NUR9>3.0.CO;2-H

Tashakkori, A., & Teddlie, C. (2010). *Handbook of mixed methods in social and behavioral research*. (2nd ed.) Thousand Oaks, CA: Sage.

Teddlie, C., & Tashakkori, A. (2003). Major issues and controversies in the use of mixed methods in the social and behavioral sciences. In A. Tashakkori & C. Teddlie (Eds.), *Handbook of mixed methods in social and behavioral research* (pp. 3–50). Thousand Oaks, CA: Sage.

Teddlie, C., & Tashakkori, A. (2009). *Foundations of mixed methods research*. Thousand Oaks, CA: Sage.

Tonkin-Crine, S., Bishop, F. L., Ellis, M., Moss-Morris, R., & Everitt, H. (2013). Exploring patients' views of a cognitive behavioral therapy-based website for the self-management of irritable bowel syndrome symptoms. *Journal of Medical Internet Research*, *15*(9), e190. doi:10.2196/jmir.2672

Wiggins, B. J. (2011). Confronting the dilemma of mixed methods. *Journal of Theoretical and Philosophical Psychology*, *31*(1), 44–60. doi:10.1037/a0022612

Yardley, L. (2001). Mixing theories: (how) can qualitative and quantitative health psychology research be combined? *Health Psychology Update*, *10*, 6–9.

Yardley, L., & Bishop, F. (2008). Mixing qualitative and quantitative methods: A pragmatic approach. In C. Willig & W. Stainton Rogers (Eds.), *The Sage handbook of qualitative research in psychology* (pp. 352–369). London, UK: Sage.

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